counting as two real ones. But recall that the vector has to be normalized as in Eq. 2.4. The normalization condition gives us one equation involving real variables, and cuts the number of parameters down to three.

As I said earlier, we will eventually see that the physical properties of a state-vector do not depend on the overall phase-factor. This means that one of the three remaining parameters is redundant, leaving only two—the same as the number of parameters we need to specify a direction in three-dimensional space. Thus, there is enough freedom in the expression to describe all the possible orientations of a spin, even though there are only two possible outcomes of an experiment along any axis.

$$\alpha_u|u\rangle + \alpha_d|d\rangle$$

2.6 REPRESENTING SPIN STATES AS COLUMN VECTORS

So far, we have been able to learn a lot by using the abstract forms of our state-vectors, that is, |*u*> and |*d*> and so forth. These abstractions help us focus on mathematical relationships without worrying about unnecessary details. However, soon we will need to perform detailed calculations on spin states, and for that we'll need to write our state-vectors in column form. Because of "phase indifference," the column representations are not unique, and we'll try to choose the simplest and most convenient ones we can find.

As usual, we'll start with |*u*> and |*d*>. We need them to have unit length, and to be mutually orthogonal. A pair of columns that satisfies these requirements is

$$|u\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \qquad (2.11)$$